

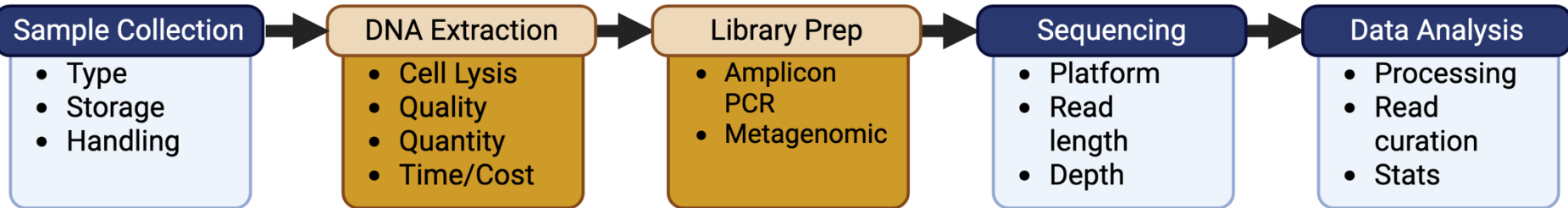
DNA Extraction & Library Preparation



Microbiome Kickstart Workshop
August 6th

Molly Bletz, Dept of Ecosystem Science & Management

Roadmap



Why it matters: DNA quality & prep impacts microbiome insights

The DNA Extraction Challenge

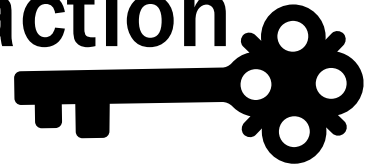
Goal: Obtain high-quality,
representative DNA from
all microbes in your
sample.

Diverse microbial cell
structures: bacteria,
fungi, archaea, viruses

Low biomass samples
and host contamination

Inhibitors: e.g., humic
acids, bile salts,
polysaccharides

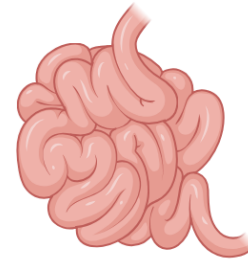
Key Considerations for DNA Extraction



- **Sample Type**
- **Cell Lysis**
- **DNA Purification/Elution**
- **Host DNA Removal**
(sometimes important)
- **Controls**

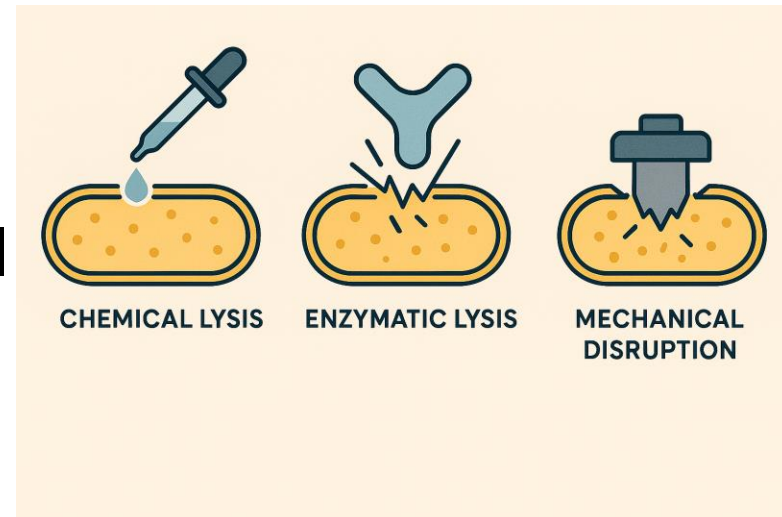
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- **Cell Lysis:** Mechanical (bead-beating), enzymatic, chemical

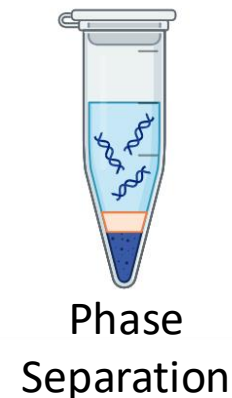
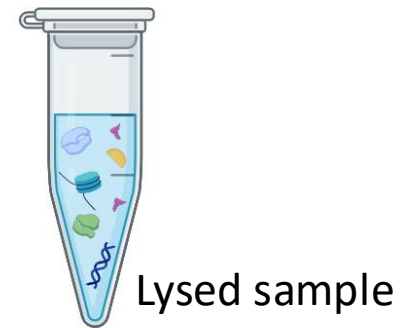


Things to ponder:

- Aggressiveness
- Duration
- Amount of sample

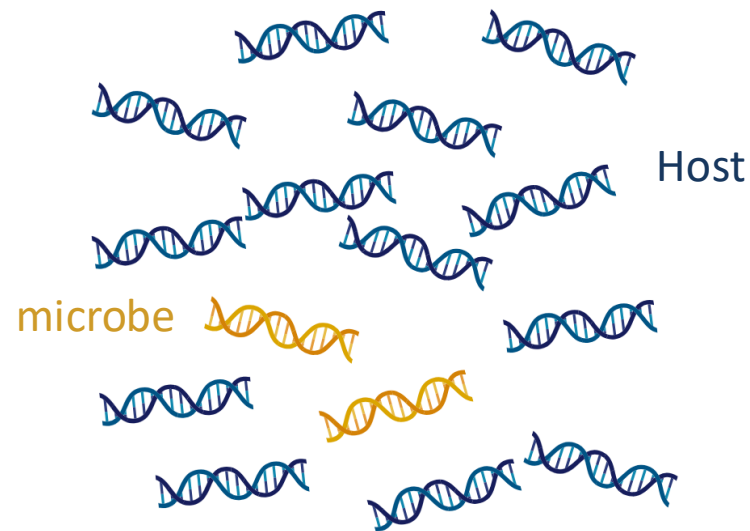
Key Considerations for DNA Extraction

- **Sample Type:** Feces, soil, tissue, swabs
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Key Considerations for DNA Extraction

- **Sample Type:** Feces, soil, tissue, swabs
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- **DNA Purification/Elution:** Silica columns, magnetic beads, phenol-chloroform
- **Host DNA Removal:** Differential lysis, NEBNext kits
- **Controls**



Common DNA Extraction Methods/Kits

- **Why Kits? Standardized, validated**
- **In-house protocols: customizable, need validation**
- **Best Practice: Match method to sample type and question**





Garbage in, Garbage out



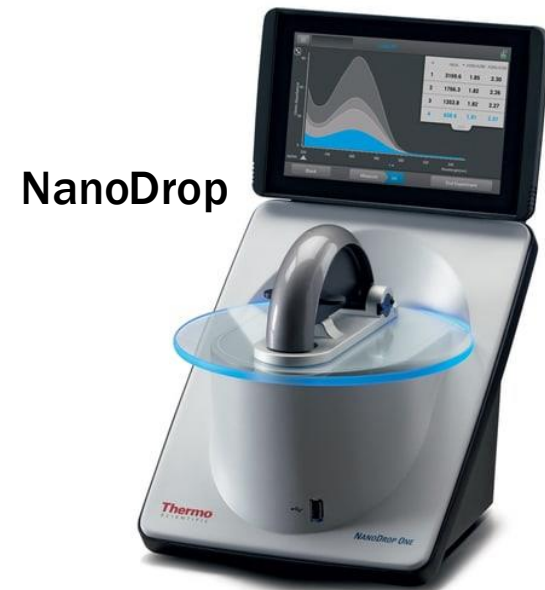
- **Goal: Clean, high molecular weight DNA**
- ***QC of DNA is key***
- **How much DNA is there?**
- **Is the DNA pure?**
- **Is the DNA intact or degraded?**



Garbage in, Garbage out



- **Goal: Clean, high molecular weight DNA**
- *Quality & Quantity of DNA is key*
- **How much DNA is there?**





Garbage in, Garbage out



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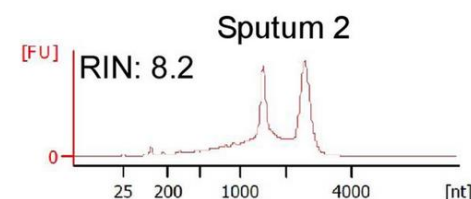
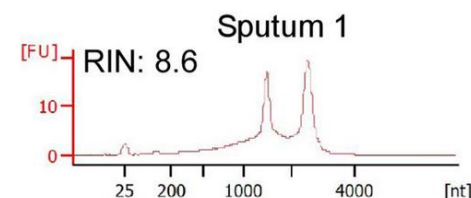
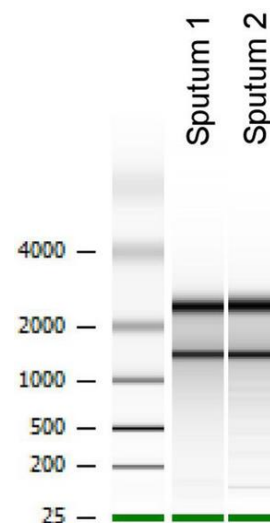
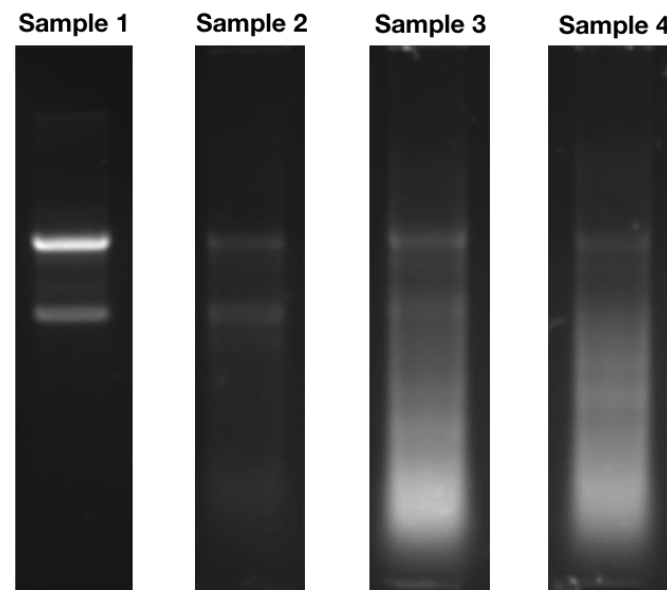
$A_{260}/_{280} \approx 1.8-2.1$,
 $A_{260}/_{230} > 1.8$



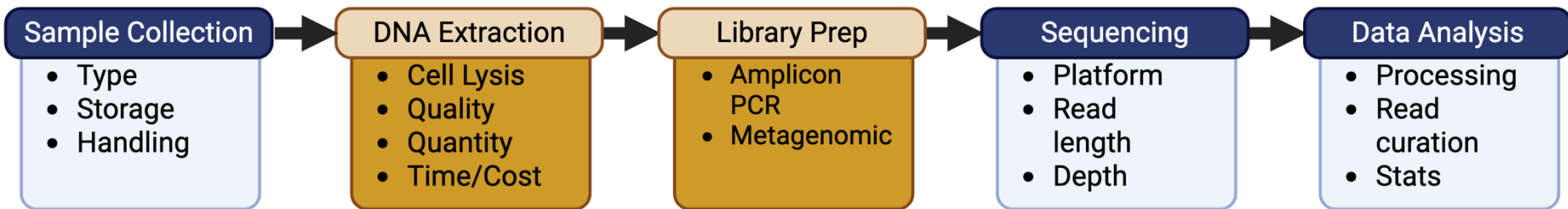
Garbage in, Garbage out



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- *Quality & Quantity of DNA is key*
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Roadmap



What are the key things to think about for DNA extraction?

Library Preparation

What

Process of preparing DNA for NGS

Why

Required for sequence platforms

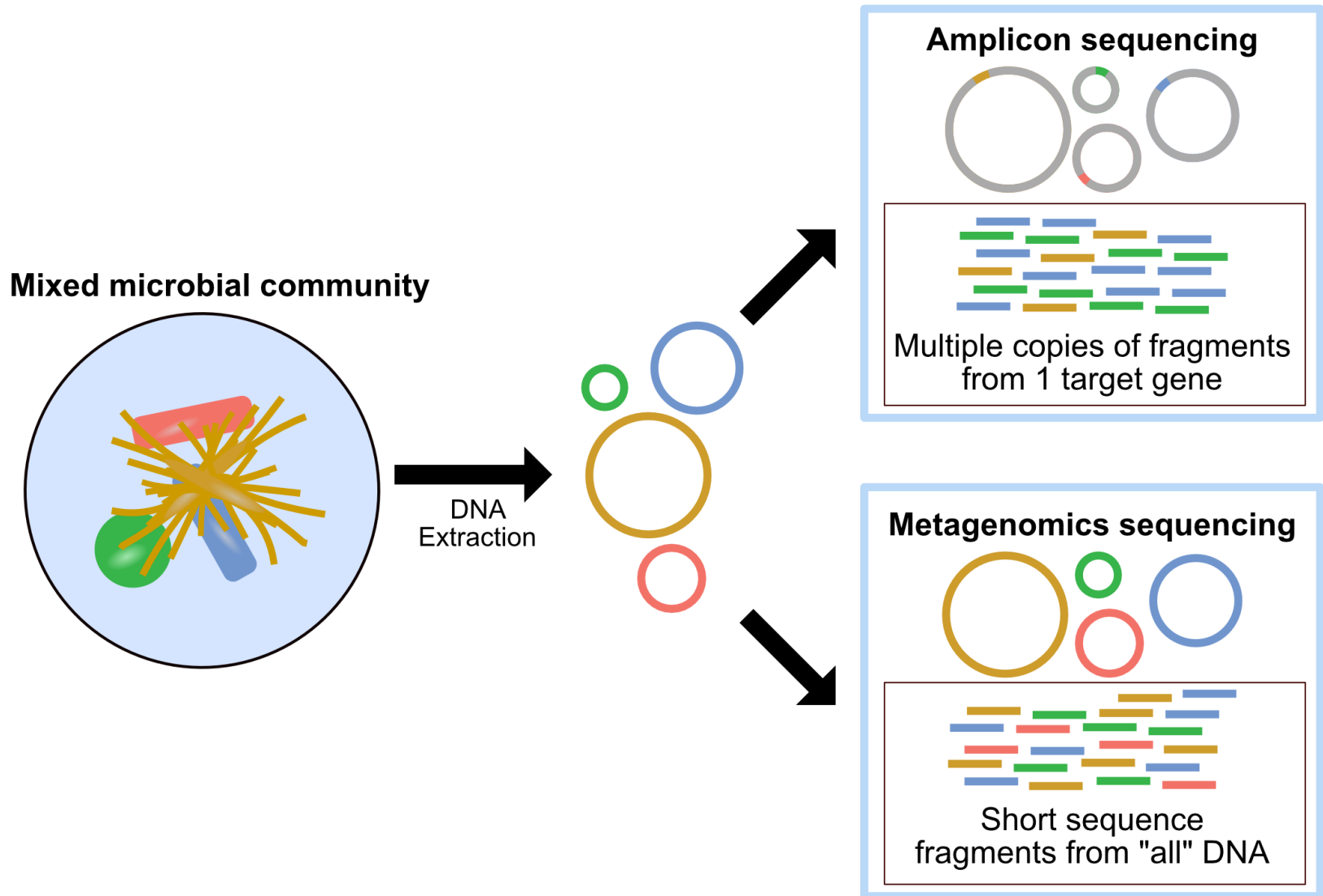
How

Two Approaches: PCR-based and Shotgun

Barcode: *unique* short DNA sequence to allow the differentiation of multiple samples

Adaptor: short DNA sequence that enables DNA fragment to attach to sequencing platform

Library Types



Library Preparation

- ✓ Fragment /Amplify DNA
- ✓ Ligate Adaptors
- ✓ Add barcodes
- ✓ Pool samples
- ✓ Clean up

PacBio

illumina®

 Oxford
NANOPORE
Technologies



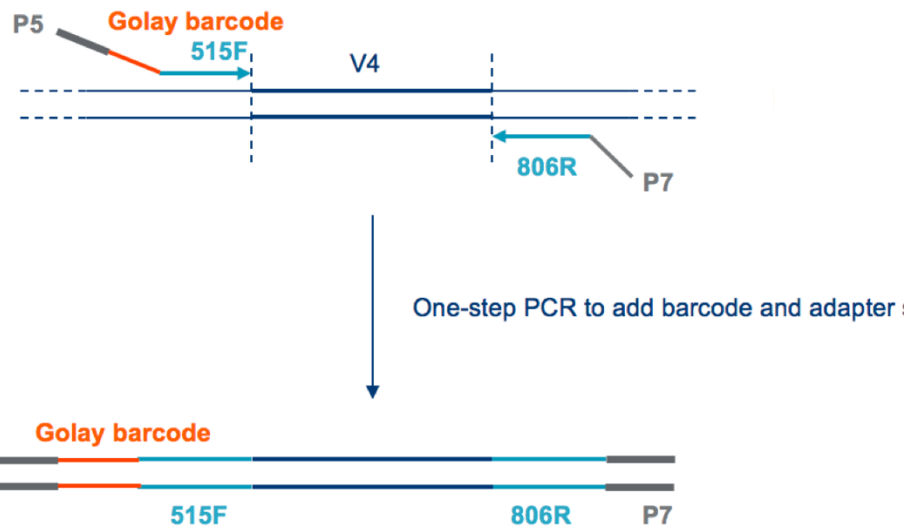
MiSeq



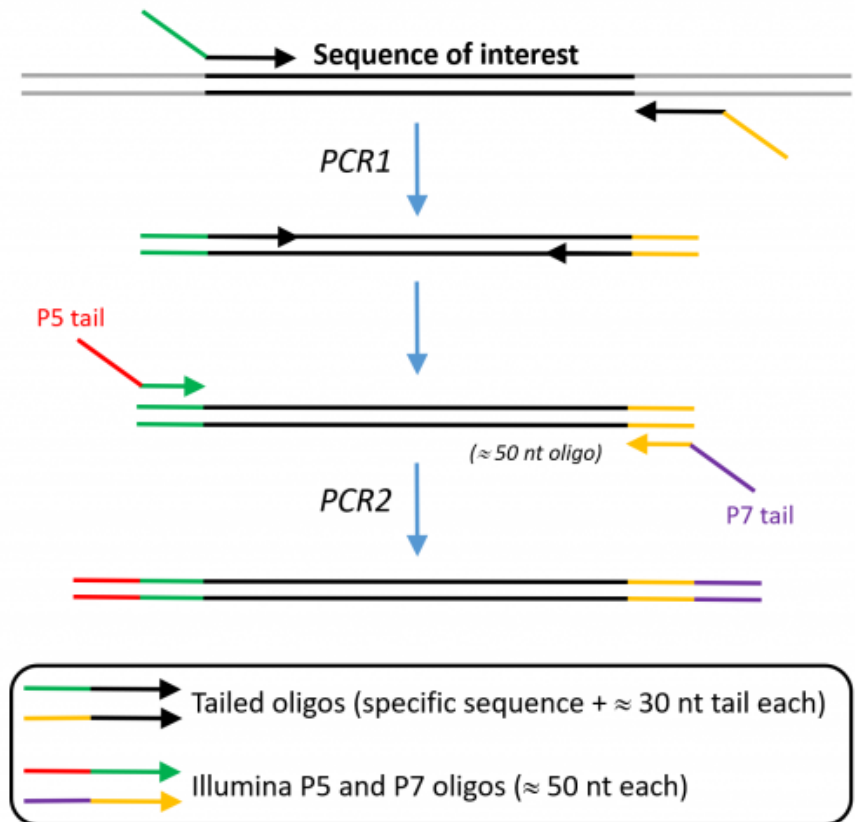
NextSeq 500

Library Prep: Amplicon Sequencing

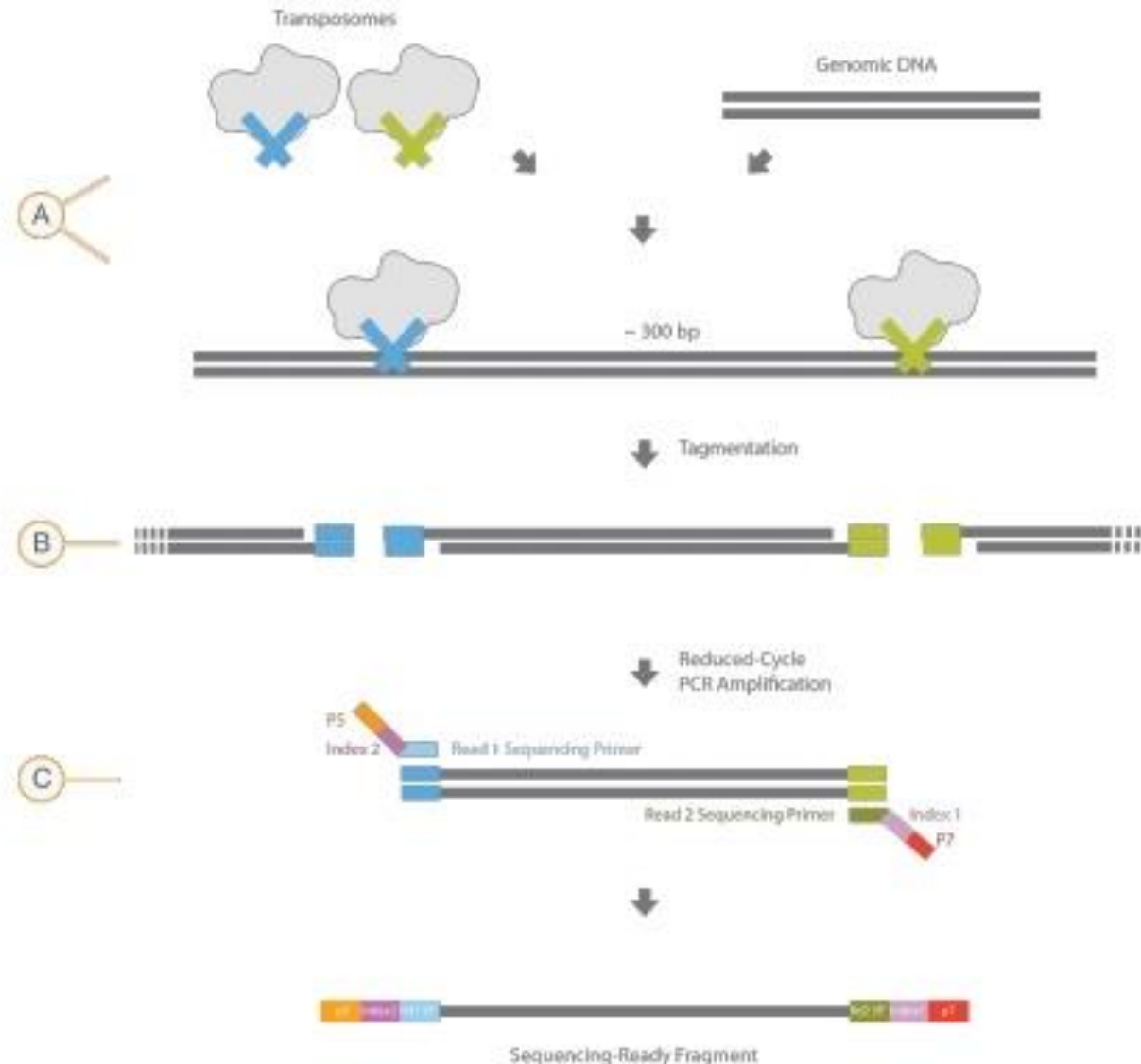
1 Step



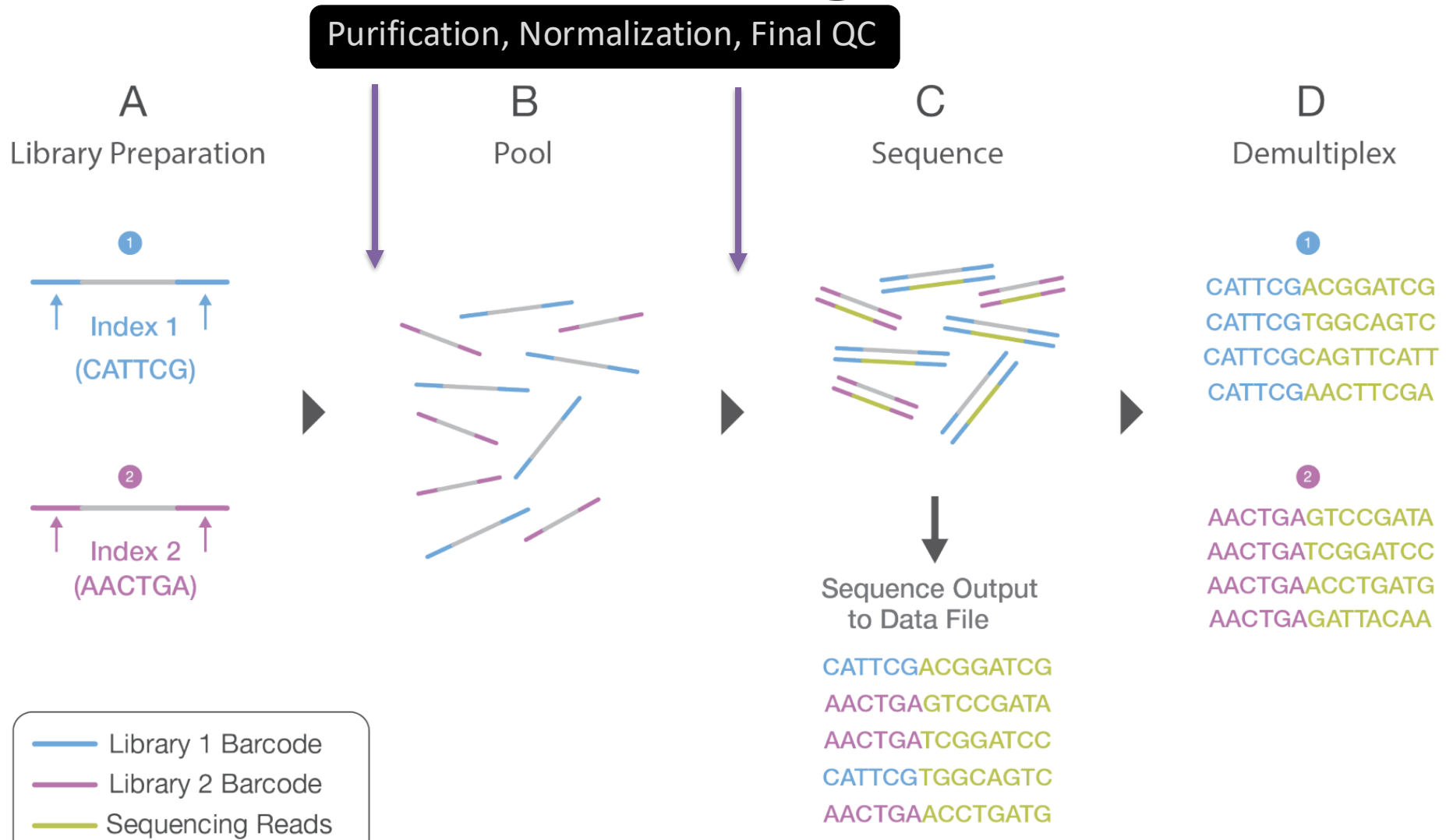
2 Step



Library Prep: Shotgun Metagenomics

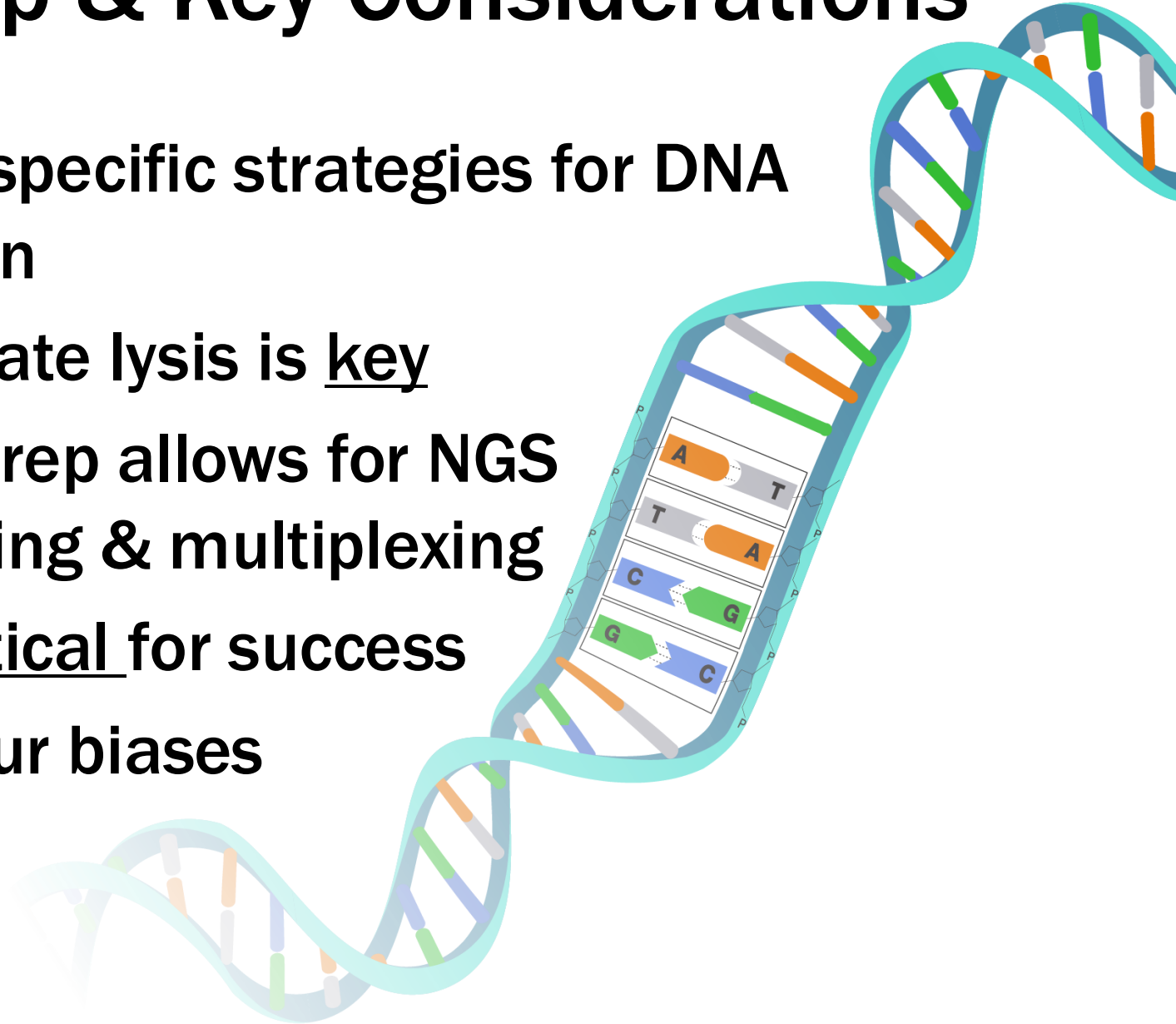


Library Preparation: Clean up & Pooling



Recap & Key Considerations

- Sample-specific strategies for DNA extraction
- Appropriate lysis is key
- Library prep allows for NGS sequencing & multiplexing
- QC is critical for success
- Know your biases



Mini-Activity: Design your methods!

- You are embarking on a new project to explore the **microbiome of amphibian skin**. Beyond bacteria, you're interested in **microeukaryotes** (particularly protists), which may play crucial roles in amphibian health and disease. You also suspect the amphibian skin might harbor **peptide compounds or toxins** that could act as inhibitors. Your goal is to identify both the protist composition and their potential interactions with the bacterial community.



Mini-Activity: Design your methods!

– DNA Extraction:

- what are the primary challenges you might anticipate for DNA extraction from **amphibian skin swabs**?
- Which **lysis methods** would you prioritize to ensure you capture DNA from *all* these diverse microbial groups, while also overcoming the challenge of inhibitors?

– Library Preparation:

- Would you choose **targeted amplicon sequencing** or **shotgun metagenomic sequencing** for this project? Why?
- What specific **QC steps** would you perform to ensure good data is generated?



Image: M. Vences